AWS – IAM – Identity and Access Management

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Purpose of IAM

The purpose of IAM (Identity Access Management) is to provide Authentication and Authorization for the Users and API requests.

Concepts of IAM Policies

- A policy states one or more permissions.
- By Default, all permissions are implicit Denied.
- An **explicit deny, overrides the explicit Allow policies.**

- There are pre-build IAM policies that can be assigned to users and Groups.
  - **Administrator Access** - Full Access to All AWS resources.
  - **Full Access** - Full Access to respective Resources,
  - **Power User Access** - Admin access, except it does not allow users/groups mgmt.
  - **Read only Access** - Only View Access to the resources.

amazon web services™

Concepts of IAM Policies

- We could also create Custom IAM policies using the **VISUAL EDITOR or write an JSON file.**
- Users and Groups can have more than one Policies attached to it.
- To Assign a Policy to an Resource, we would need to create **ROLES**.
- There is an option to assign an Condition to the custom Policy as well.

Components in IAM

- Users – Authentication to Individual person
- Groups – Group of Users using the same Policy
- Policy -- Set of Rules that is permitted or Denied for a Group or User
- Roles -- Authorizing “AWS” to do a function or an Action of behalf of us by assigning an Policy to it.
- Access / Secret Access Keys – Used for API’s and CLI based Access.

Policy

- Each policy has two type of Editing, “Visual Editor” and “JSON” script.
- Under policy Script, there are 3 parts.
  - Action
  - Effect
  - Resource.

Policy

- **Action** → What feature or component on which you want to perform the Action. Eg: -- EC2, S3, elasticLoadbalancing, Autoscaling,
- **SubAction** → Under a particular Feature, what all we want Allow or Deny.
  - Eg: -- EC2:describeinstances
- https://docs.aws.amazon.com/IAM/latest/UserGuide/reference_policies_actions-resources-contextkeys.html
- The above link refers to all the types of Action's and Sub Action's in AWS IAM.
- For Every Action in the features, there are 3 parts of it
  - List
  - Read
  - Write

Policy

- **Effect** → to Allow or Deny the Actions.
- **Resource** → What particular Resource should be actioned.
  - Eg:-- "arn:aws:s3:::awsb15we“
  - Eg:-- arn:${Partition}:ec2:${Region}:${Account}:customer-gateway/${CustomerGatewayId}

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User Creation

- **Programmatic Access** used for AWS API, CLI and SDK's and works on
  - Access Key ID
  - Secret Access key
- While creating the user, AWS creates an one time show of "Secret Access key" similar to your private keys.
- **Console Access** used to login to AWS via GUI works on
  - Username
  - Password
- For GUI user access the below link is used.
  - https://signin.aws.amazon.com/console
  - We need to provide, Account number, Username and Password
- While creation a users, 3 options are
  - Assign it to a Group
  - Copy the Existing permission from one of the user
  - Assign the policy on the user Directly.

User -- Permission

3 Options

- Can assign the users to the Group
- Can copy the existing user permissions to this User
- Can assign the policy Directly to the User

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AWS CLI Access

- Re-Requisite
- Install the AWS CLI software on Windows or Linux. (your OS)
- https://docs.aws.amazon.com/cli/latest/userguide/awscli-install-windows.html
- For Login → open "Commond Prompt" an then run the below command
- **"aws configure"**
- You have to provide 4 information
- 1. Access key
- 2. Secret Access key
- 3. Region
- 4. Display type → "**text**", "**table**" or "**json**"

Region Codes for CLI

Region Name	Region	Endpoint	Protocol	Amazon Route 53 Hosted Zone ID*
US East (Ohio)	us-east-2	apigateway.us-east-2.amazonaws.com	HTTPS	ZOJJZC49E0EPZ
US East (N. Virginia)	us-east-1	apigateway.us-east-1.amazonaws.com	HTTPS	Z1UJRXOUMOOFQ8
US West (N. California)	us-west-1	apigateway.us-west-1.amazonaws.com	HTTPS	Z2MUQ32089INYE
US West (Oregon)	us-west-2	apigateway.us-west-2.amazonaws.com	HTTPS	Z2OJLYMUO9EFXC
Asia Pacific (Mumbai)	ap-south-1	apigateway.ap-south-1.amazonaws.com	HTTPS	Z3VO1THU9YC4UR
Asia Pacific (Seoul)	ap-northeast-2	apigateway.ap-northeast-2.amazonaws.com	HTTPS	Z20JF4UZKIW1U8
Asia Pacific (Osaka-Local)**	ap-northeast-3	apigateway.ap-northeast-3.amazonaws.com	HTTPS	Z2YQB5RD63NC85
Asia Pacific (Singapore)	ap-southeast-1	apigateway.ap-southeast-1.amazonaws.com	HTTPS	ZL327KTPIQFUL
Asia Pacific (Sydney)	ap-southeast-2	apigateway.ap-southeast-2.amazonaws.com	HTTPS	Z2RPCDW04V8134
Asia Pacific (Tokyo)	ap-northeast-1	apigateway.ap-northeast-1.amazonaws.com	HTTPS	Z1YSHQZHG15GKL
Canada (Central)	ca-central-1	apigateway.ca-central-1.amazonaws.com	HTTPS	Z19DQILCV0OWEC
China (Beijing)	cn-north-1	apigateway.cn-north-1.amazonaws.com.cn	HTTPS	None
China (Ningxia)	cn-northwest-1	apigateway.cn-northwest-1.amazonaws.com.cn	HTTPS	None
EU (Frankfurt)	eu-central-1	apigateway.eu-central-1.amazonaws.com	HTTPS	Z1U9ULNL0V5AJ3
EU (Ireland)	eu-west-1	apigateway.eu-west-1.amazonaws.com	HTTPS	ZLY8HYME6SFDD
EU (London)	eu-west-2	apigateway.eu-west-2.amazonaws.com	HTTPS	ZJ5UAJN8Y3Z2Q
EU (Paris)	eu-west-3	apigateway.eu-west-3.amazonaws.com	HTTPS	Z3KY65QIEKYHQQ
South America (São Paulo)	sa-east-1	apigateway.sa-east-1.amazonaws.com	HTTPS	ZCMLWB8V5SYIT

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CLI command for IAM features

IAM – list users

```
[c:\~]$ aws iam list-users
-----------------------------------------------------------------------------------------------------------------------------------
|                                                ListUsers                                                                          |
+-----------------------------------------------------------------------------------------------------------------------------------+
||                                                Users                                                                            ||
|+-------------------------------------+----------------------+----------------------+-------+----------------------+------------+|
||            Arn                      |     CreateDate       |  PasswordLastUsed    | Path  |      UserId          | UserName   ||
|+-------------------------------------+----------------------+----------------------+-------+----------------------+------------+|
|| arn:aws:iam::766476266373:user/user1 | 2018-08-03T15:18:37Z | 2018-08-03T15:21:29Z | /     | AIDAJCVJUE7PQRZXQJLW4 | user1      ||
|| arn:aws:iam::766476266373:user/user2 | 2018-08-04T06:45:36Z | 2018-08-04T06:49:15Z | /     | AIDAIJHEPYVWEQJ5PLCNY | user2      ||
|+-------------------------------------+----------------------+----------------------+-------+----------------------+------------+|
```

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